

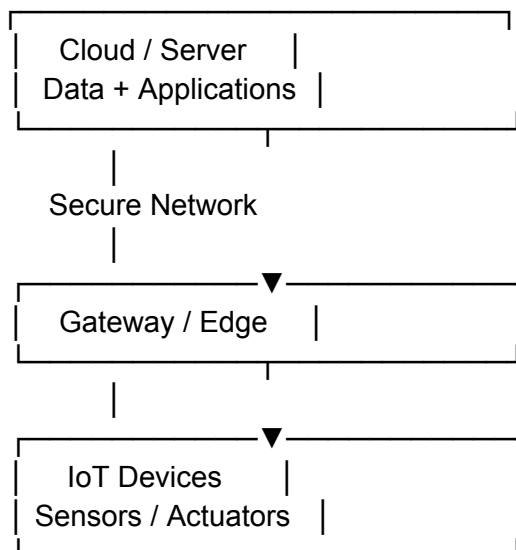
IoT Security and Threats

1. Introduction to IoT Security

IoT Security refers to the techniques, technologies, policies, and practices used to **protect IoT devices, communication networks, data, and cloud services from unauthorized access, attacks, damage, and misuse.**

IoT security is challenging because IoT systems contain a very large number of **heterogeneous, resource-constrained, and often physically accessible devices** that continuously communicate over networks.

Basic IoT Security Architecture

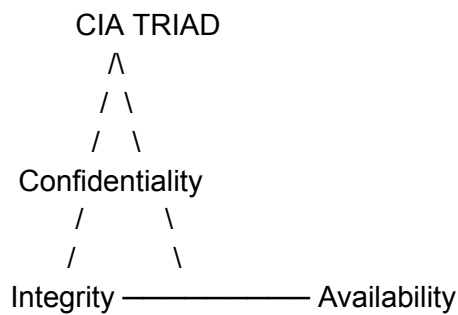


Security must be maintained across **all these layers**, not just at the device level.

2. CIA Triad in IoT Security

The **CIA Triad** is one of the fundamental security models used to protect information and systems.

CIA = Confidentiality + Integrity + Availability



1. Confidentiality

Confidentiality ensures that IoT data is accessible **only to authorized users or devices**.

Example:

Patient data collected by healthcare IoT devices should not be accessible to unauthorized persons.

Security mechanisms:

- Encryption
 - Authentication
 - Access control
-

2. Integrity

Integrity ensures that IoT data is **not altered, modified, or corrupted without authorization**.

Example:

An attacker should not be able to modify a smart meter's energy reading.

Security mechanisms:

- Hashing
 - Digital signatures
 - Message authentication codes
-

3. Availability

Availability ensures that IoT devices, services, and data are **available to authorized users whenever required**.

Example:

A smart security system must remain operational when an emergency occurs.

Security mechanisms:

- Backup systems
- Redundancy
- DDoS protection
- Fault tolerance

★ Easy Memory

Confidentiality → Keep it **SECRET**

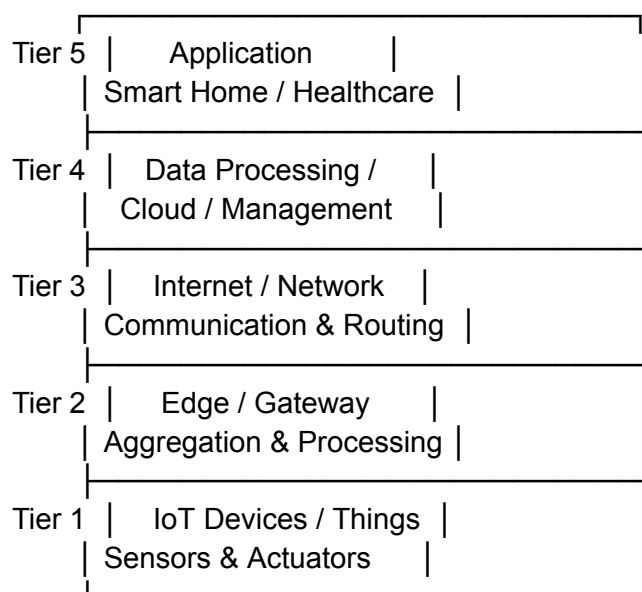
Integrity → Keep it **CORRECT**

Availability → Keep it **ACCESSIBLE**

3. Tiered IoT Architecture for Security

IoT systems are commonly represented using a **tiered/layered architecture** to understand where security threats and protections apply.

A practical **five-tier IoT architecture** can be represented as:



Security at Different Tiers

Tier	Main Component	Major Security Concern
1	Sensors & Actuators	Physical attacks, device compromise
2	Edge/Gateway	Unauthorized access, malware
3	Network	Eavesdropping, spoofing, DoS
4	Cloud/Data Processing	Data breaches, unauthorized access
5	Application	Account attacks, privacy violations

Important Point

Security should be **end-to-end**. Protecting only the cloud while leaving IoT devices insecure can still allow an attacker to compromise the entire system.

4. Common IoT Security Threats

IoT systems face threats at the **device, network, communication, cloud, and application levels**.

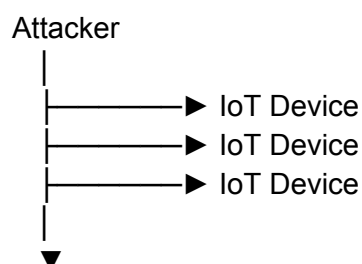
1. Malware

Attackers can infect IoT devices with malicious software.

Example: A compromised IoT device may become part of a **botnet**.

2. Denial of Service (DoS / DDoS)

The attacker sends a large number of requests or packets to make an IoT service **unavailable**.



Service Overloaded



Unavailable

3. Eavesdropping

An attacker secretly **intercepts communication** between IoT devices.

Example:



Encryption helps protect against this threat.

4. Spoofing

An attacker **pretends to be a legitimate device or user**.

Example: A fake sensor sends false temperature readings to the IoT network.

5. Man-in-the-Middle (MITM)

The attacker positions themselves between two communicating devices and may **intercept or modify messages**.



6. Physical Attacks

IoT devices are often deployed in public or physically accessible locations.

An attacker may:

- Physically damage the device.
- Steal the device.
- Extract stored information.

- Tamper with hardware.
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7. Weak Authentication

Default, weak, or reused passwords can allow attackers to gain unauthorized access.

8. Insecure Firmware

Outdated or vulnerable firmware can contain security weaknesses that attackers can exploit.

9. Data Privacy Threats

IoT devices can collect sensitive information such as:

- Location
- Health information
- Home activity
- User behavior

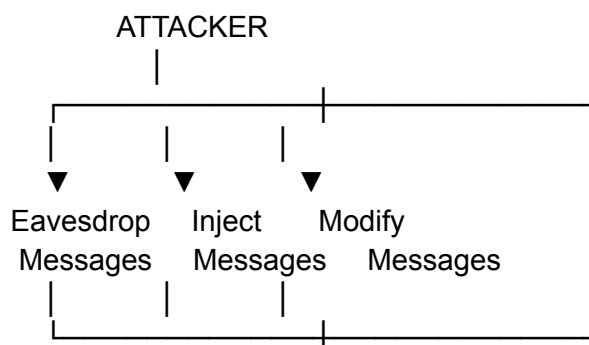
Improper protection can result in **privacy violations and data leakage**.

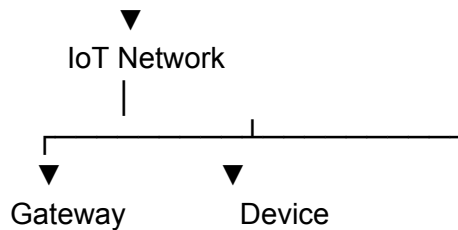
5. IETF Attacker Model

The **IETF attacker model** is used to describe the capabilities and position of an attacker in a networked/IoT environment.

In security analysis, an attacker may be modeled based on **what they can access, observe, modify, inject, or control**.

A useful representation is:





Major Attacker Capabilities

1. Eavesdropping

The attacker can **listen to network traffic** and obtain transmitted information.

2. Message Injection

The attacker can **insert fake messages** into the communication system.

3. Message Modification

The attacker can **alter messages while they are being transmitted**.

4. Message Replay

The attacker captures a legitimate message and **retransmits it later** to fool the receiver.

5. Device Compromise

The attacker gains control of an IoT device and uses it to attack other devices or services.

6. Denial of Service

The attacker attempts to prevent legitimate users or devices from accessing services.

★ Important Exam Concept

The attacker model helps security designers answer:

"What can the attacker see, control, modify, inject, or disrupt?"

Based on these capabilities, appropriate security mechanisms such as **encryption, authentication, integrity checks, access control, secure boot, and intrusion detection** can be applied.

6. IoT Security Measures

To protect IoT systems, the following measures can be used:

Device Level

- Strong authentication
- Secure boot
- Firmware updates
- Device identity management
- Physical security

Network Level

- Encryption
- Firewalls
- Network segmentation
- Intrusion detection
- Secure communication protocols

Data Level

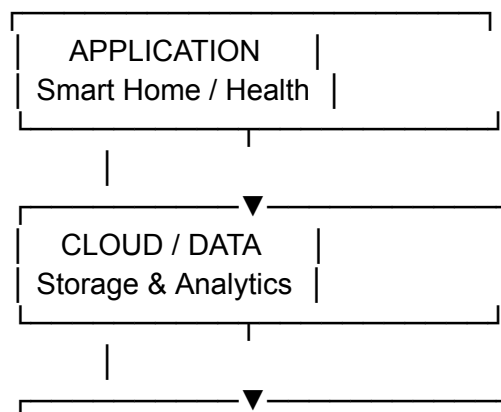
- Encryption at rest and in transit
- Hashing
- Access control
- Data backup

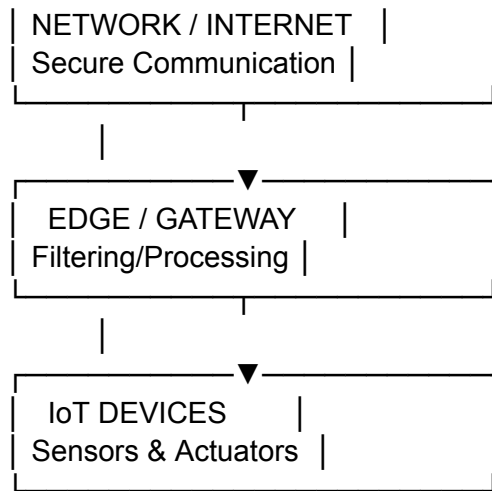
Application/Cloud Level

- Multi-factor authentication
- Authorization
- Secure APIs

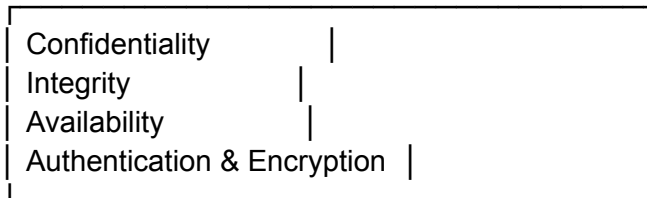
- Regular security monitoring
- Vulnerability management

Complete Exam Diagram





SECURITY ACROSS ALL TIERS



Possible Attacker Actions

